# AYMANE AIT DADS

KI-Entwickler Werkstudent | Computer Vision · Image Recognition · Web-App Integration

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## PROFESSIONAL SUMMARY

Data Science engineering student with hands-on experience training and deploying image recognition models, including a CLIP ViT-L/14-based classifier ranked 1st in class (0.791 AUC) across 250K images from 25+ generator types, directly matching XRbit's computer vision and object recognition requirements. At Orange Maroc, built end-to-end ML pipelines on 100K+ records using Python and Scikit-learn, reducing manual analysis time by ~60% and delivering stakeholder-ready outputs via Power BI. Proficient in PyTorch, TensorFlow, feature engineering, dataset construction, REST APIs, Supabase (PostgreSQL), and Docker, covering the full stack from AI model training to web-app database integration. Available for Werkstudent engagement in Europe; no work authorization restrictions apply.

## CORE COMPETENCIES

Image Recognition | Computer Vision | KI-Modell Training | Datensatz-Optimierung | Web-App Entwicklung | API & Datenbank-Anbindung | Objekterkennung | Systemarchitektur-Integration

## WORK EXPERIENCE

### Orange Maroc
*Jul 2024 - Aug 2024*

**Data Science Intern** - Casablanca, Morocco

- Built and optimized an end-to-end ML pipeline combining Random Forest and K-Means on **100K+ fiber-optic network records** (upload speed, latency, jitter) to classify network quality across thousands of nodes.
- Developed a clustering pipeline mapping customer performance profiles to **5 commercial service tiers**, with output adopted directly by the network optimization team for production decision-making.
- Reduced manual analysis time by **~60%** through automated feature engineering, dataset construction, and systematic model evaluation using Python, Scikit-learn, and Pandas.
- Delivered stakeholder presentations translating AI model outputs into actionable maintenance recommendations, visualized through **Power BI** dashboards for non-technical audiences.

## PROJECTS

**AI-Generated Image Detection - NTIRE 2026 @ CVPR + EURECOM ImSecu** *EURECOM Course + International Benchmark · 2025-2026 · Ranked 1st in Class*

- Trained a CLIP ViT-L/14 (428M params) backbone with a custom MLP head on **250K images from 25+ generator types**, achieving 0.791 AUC and ranking 1st on the EURECOM private Kaggle class leaderboard; separately submitted to NTIRE 2026 @ CVPR CodaBench international benchmark.
- Applied dual learning-rate fine-tuning (2e-6 for CLIP, 5e-4 for head), 10-view TTA (flips, crops, blur, resizes), and 3-model ensemble weighted by validation AUC, with ablation studies confirming **1-epoch training outperforms 4-epoch** for out-of-distribution generalization.

PyTorch · OpenCLIP · ViT · Kaggle T4 · FP16 Mixed Precision

**Aerial Cactus Object Detection - Endangered Species Classification** *EURECOM · 2025 · 99.8% Accuracy*

- Benchmarked CNN, DenseNet121, and a hybrid CNN-Transformer architecture on **17,500+ drone images** for endangered cactus species classification, achieving 99.8% accuracy, F1 = 0.999, and ROC AUC = 1.0.
- Constructed and optimized the full dataset pipeline - from raw image ingestion to model evaluation - demonstrating end-to-end **image recognition system integration** matching real-world object detection requirements.

PyTorch · DenseNet121 · CNN · Transformer · Scikit-learn

## EDUCATION

**Engineering Degree in Data Science (Master-level) -** EURECOM *Sep 2023 - Present*

Relevant coursework: Machine Learning, Deep Learning, Computer Vision, Image Security, NLP, Cloud Computing, Web Applications, Statistics

**CPGE - Mathematics & Physics -** Ibn Timiya *2021 - 2023*

## SKILLS

**Computer Vision & AI Model Training:** PyTorch | TensorFlow | Scikit-learn | OpenCLIP | ViT / CNN / DenseNet | Image Classification | Objekterkennung

**Web-App, API & Datenbank-Integration:** REST APIs | Supabase | PostgreSQL | Docker | Python | Bash | Power BI

**MLOps & Datensatz-Optimierung:** Feature Engineering | Ablation Studies | FP16 Mixed Precision | Git | Linux | Pandas | NumPy